

Development of KR-EEIO & GHG-Simulator

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1.1 Overview

Environmentally Extended Input-Output (EEIO)

- Sub-item of UN's System of Environmental-Economic Accounting(SEEA)
 - o Compiled by combining environmental statistics(GHG, energy, water) and IOTs (GHG only in KR-EEIO)
- Basic data for environmental and climate-related analysis
- Enables Scope3 calculation by measuring indirect emissions
- Provides industry-average coefficients for rough calculations for SMEs that have difficulty measuring direct GHG emissions (serves as baseline data for GHG-Simulator)

1.1 Overview

T mat GHG(Total Table, incl. GHG)

Products	products	Agriculture	Manufacturing	Services	FD	Total Demand	Total Output	Imports
Agriculture		Intermediate Input						
Manufacturing								
Services								
VA		VA by Industry						
Total Input		Total Input by Industry						
GHG(t co2eq)		GHG by Industry						

- Same format as a standard IO table, but adds a GHG emissions (t CO2eq) row at the bottom, calculated via the estimation of GHG emissions by Industry
- Target year : 2015, 2020~2023
- Industry Sectors : Integrated Large Classification(33 Sectors)
- Prices : Current and Constant prices

1.1 Overview

Components

Items	Variable Name	Description
Total table	T_mat_GHG	IO Table (Import + Domestic) and emissions by industry
Input(GHG) coefficient	Ad	Emissions per unit of output (KRW million)
Production Inducement Coefficient	Lf	Total industry output induced by final demand (KRW million) for a specific product
Emission Inducement Coefficient	M	Total industry emissions induced by final demand (KRW million) for a specific product
Emissions by Final Demand Item	M-fd	Total emissions induced by each final demand item
Emissions based on Production Value	M·X(diag)	Total industry emissions induced by the production value of a specific product
Emission Scope	Scopes 1, 2, 3	Direct/Indirect emission classification
Import Transactions Table	I_mat	IO Table (Import)
Domestic Transactions Table	D_mat	IO Table (Domestic)

- Calculated using GHG by industry and IOTs, produced in both current and constant prices

1.2 Estimation of GHG Emissions by Industry

Basic Data

- National Inventory Report(NIR, 7 GHGs)
- Extended Energy Balance(EEB)
- IOTs(Current price published by Input-Output Team, Constant price additionally compiled)

Compilation procedure

- ① Match the industry classification of NIR and IOTs
- ② Calculate GHG emission by energy source using EEB
- ③ Allocate emissions calculated in step2 by industry using IOTs

1.2 Estimation of GHG Emissions by Industry

① NIR and IOTs Industry Classification Matching

<GHG Inventory>
(kt co2eq)

Sector / Category / Year	2023	
Total Emissions	707,199.78	
Net Emissions	667,766.58	
1. Energy	544,988.30	
A. Fuel Combustion	541,746.31	
1. Energy Industries	248,155.39	
a. Public Electricity and Heat Production	226,959.86	D
b. Petroleum Refining	19,605.15	C04
c. Manufacture of Solid Fuels & Other Energy Ind.	1,590.37	B, C04, D
2. Manufacturing Industries and Construction	147,240.38	
a. Iron and Steel	74,997.26	C07
b. Non-Ferrous Metals	3,775.43	C07
c. Chemicals	24,925.76	C05
d. Pulp, Paper and Print	2,750.24	C03
e. Food Processing and Tobacco	2,065.05	C01
f. Non-Metallic Minerals	15,432.97	C04
g. Other	23,293.66	
1. Machinery	4,801.41	C11
2. Transport Equipment	2,362.19	C12
3. Mining	165.60	B
4. Wood and Wood Products	93.51	C03
5. Construction	1,813.02	F
6. Textiles and Leather	4,524.77	C02
8. Other	9,533.15	C06, C08, C09, C10, C13, C14
3. Transport	97,825.94	
a. Civil Aviation	1,771.69	H
b. Road Transportation	94,414.14	H, 9111
c. Railways	182.56	H
d. Domestic Navigation	1,383.41	H
e. Other Transport	74.15	H
4. Other	45,502.68	
a. Commercial / Institutional	13,988.28	All services except transport
b. Residential	27,043.36	9111
c. Agriculture / Forestry / Fishing	3,671.04	A
Agriculture / Forestry	1,311.25	
Fishing	2,359.78	
5. Unclassified	3,021.92	O
B. Fugitive	3,241.99	
1. Solid Fuels	2,334.02	D
2. Oil and Natural Gas	907.96	B

- One-to-one matching : No additional work required
- One-to-many matching : Additional work (using EEB) required (total 4 items)

Allocate by industry via CO2-equivalent calculated by energy source

Calculate CO2-equivalent by energy source, then allocate by industry using the industry-linkage table

1.2 Estimation of GHG Emissions by Industry

② Calculating GHG Emissions by Energy Source Using the Extended Energy Balance

- For each energy source, multiply calorific value(TJ) x carbon coefficient x CO2 conversion factor(44/12) to calculate the CO2-equivalent amount, then match with the IO code

< CO2-Equivalent Calculation >

EEB(A)	Sector	2023															
		Domestic anthracite	Imported anthracite	Coking coal	Other bituminous coal	Sub-bituminous coal	Lignite	Peat	Solid fuel	Coke	Coal tar	Coke oven gas	Blast furnace gas	Other coal gas	Natural gas	City gas	Crude oil
	(TJ)																
	Other manufacturing	0	61686	0	0	0	0	0	29	0	0	0	0	0	9842	5324	0
	Road	0	0	0	0	0	0	0	0	0	0	0	0	0	0	41359	0

Carbon coefficient(B)	29.705	27.32	25.349	26.105	26.868	26.868	28.9	26.105	26.192	28.349	28.307	56.1	44.4	15.281	15.236	20
CO2 conversion factor©	3.67	3.67	3.67	3.67	3.67	3.67	3.67	3.67	3.67	3.67	3.67	3.67	3.67	3.67	3.67	3.67

CO2-equivalent (t co2eq)	Other manufacturing	0.0	6,179,262.2	0.0	0.0	0.0	0.0	0.0	2,766.3	0.0	0.0	0.0	0.0	0.0	551,433.7	297,404.7	0.0
	Road	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2,310,551.1	0.0
	Commercial	0.0	0.0	0.0	0.0	0.0	0.0	0.0	4,939.1	0.0	0.0	0.0	0.0	0.0	0.0	8,104,691.2	0.0
	Public	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	139,942.7	0.0

IO	IOcode	0611	0611	0612	0612	0612	0612	0612	0612	1610	1610	4610	4610	4610	0622	4610	0621
	IOcode name	Anthracite	Anthracite	Bituminous coal	Bituminous coal	Bituminous coal	Bituminous coal	Bituminous coal	Bituminous coal	Coal product	Coal product	City gas	City gas	City gas	Natural gas	City gas	Crude oil

1.2 Estimation of GHG Emissions by Industry

(A.c. Solid Fuel Manufacturing and Other Energy Industries)

- CO2-equivalent total is used as a ratio to allocate to B(Mining products), C04(Coal and petroleum products), D(Electricity, gas and steam)

<CO2-Equivalent for Industry Allocation Ratio>

IO code	Energy flow(2)(EEB)	Total CO2 equivalent
D	Gas manufacturing	-170,341.7
B	Coal mining	-44,773.3
B	Crude oil and natural gas extraction	0.0
C04	Briquette manufacturing_own consumption	0.0
D	Gas manufacturing_own consumption	-513,968.9
D	Synthetic coal gas manufacturing_own consumption	0.0
C04	Other conversion_own consumption	-1,137,941.7



	Total CO2 equivalent	Ratio	Distribution (NIR total)
			1590.37
B (Mining products)	44,773	2.4%	38.1
C04 (Coal and petroleum products)	1,137,942	60.9%	969.3
D (Electricity, gas and steam)	684,311	36.7%	582.9

1.2 Estimation of GHG Emissions by Industry

(A.2.g.8, Other, A.3.b. Road Transport, A.4.a. Commercial/Public)

- Allocate the calculated CO2-equivalent amount using the industry-linkage table(380x33)

Total	IO Code	Fuel	A Agriculture, Forestry & Fisheries Products	B Mining Products	C01 Food & Beverage Products	C02 Textile & Leather Products	C03 Wood, Paper & Printing	C04 Coal & Petroleum Products	C05 Chemical Products	C06 Non-metallic Mineral Products	C07 Primary Metal Products	C08 Fabricated Metal Products	C09 Computer, Electronic & Optical Equipment	C10 Electrical Equipment	C11 Machinery & Equipment
6,179,262.2	0611	Anthracite	0	223	0	0	0	30,812	30,275	98,117	936,803	58	0	0	0
2,766.3	0612	Bituminous Coal	0	0	0	0	0	71,128	98,650	710,981	6,975,842	11	0	0	0
551,433.7	0622	Natural Gas (LNG)	0	0	0	0	0	0	0	0	0	0	0	0	0
14,712.3	1622	Gasoline	78,783	8,168	79,067	29,555	46,193	174,396	157,497	24,788	28,284	84,114	48,891	28,309	76,501
123,635.5	1624	Kerosene	364,074	1,989	25,271	7,297	9,733	27,577	29,705	19,846	12,608	19,917	4,686	4,438	10,146
372,854.3	1625	Diesel	982,542	66,172	84,965	15,054	32,456	103,961	140,696	156,673	44,133	77,528	24,463	18,497	39,462
36,864	1626	Heavy Fuel Oil	73,717	26,037	152,579	110,953	150,130	2,630,966	603,047	263,428	95,868	35,969	22,655	23,117	27,129
1,789,252.2	1627	LPG	3,439	3,284	17,531	5,333	7,674	20,773	2,253,517	40,691	19,252	46,796	11,785	14,486	16,855
297,404.7	4610	City Gas	582	3,022	821,342	164,448	261,019	574,956	1,385,047	466,485	2,350,272	368,646	665,333	221,078	110,599



Using the industry allocation ratio by energy source in the industry-linkage table

Total CO2 equivalent	IO Code	Fuel	C06 Non-metallic Mineral Products	C08 Fabricated Metal Products	C09 Computer, Electronic & Optical Equipment	C10 Electrical Equipment	C13 Other Manufacturing Products	C14 Manufacturing Processing & Industrial Equipment Repair
6,179,262.2	0611	Anthracite	6,175,108.4	3,650.3	0	0	0	503.5
2,766.3	0612	Bituminous Coal	2,766.2	0	0	0	0	0
551,433.7	0622	Natural Gas (LNG)	0	0	0	0	0	0
14,712.3	1622	Gasoline	1,479.2	5,019.3	2,917.5	1,689.3	495.6	3,111.4
123,635.5	1624	Kerosene	40,456.2	40,601	9,552.4	9,046.9	5,000.5	18,978.5
372,854.3	1625	Diesel	149,108.4	73,784.7	23,281.9	17,603.9	12,440.8	96,634.5
36,864	1626	Heavy Fuel Oil	12,880.9	1,758.8	1,107.8	1,130.4	500.9	19,485.3
1,789,252.2	1627	LPG	132,646.2	152,547.6	38,417.2	47,222.1	32,500.6	1,385,918.5
297,404.7	4610	City Gas	80,022	63,238.4	114,132.9	37,924.3	2,028.5	58.7

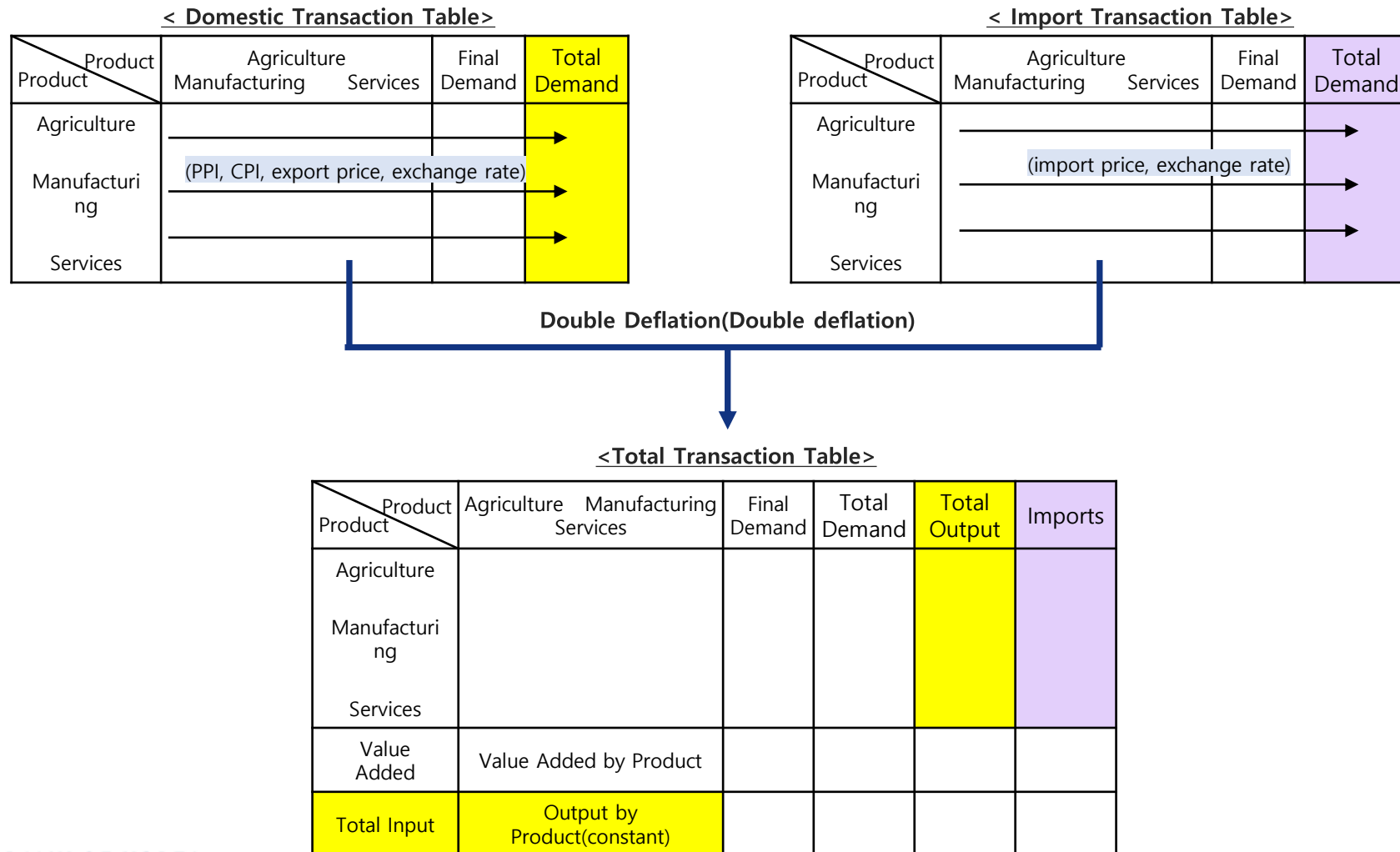
1.2 Estimation of GHG Emissions by Industry

< GHG Emissions by Industry>

Code	Sector Name	2015	2016	2017	2018	2019	2020	2021	2022	2023
A	Agriculture, Forestry & Fisheries Products	28,555.1	27,850.4	27,821.7	27,415.3	27,223.8	27,522	27,013.3	26,549.1	26,208.4
B	Mining Products	999.8	1,020.9	1,073.9	1,135.5	1,089	1,079.4	1,118	1,151.5	2,042.9
C01	Food & Beverage Products	1,915.7	2,194.4	2,089.2	2,170.3	1,928.1	2,045.8	2,111.8	2,218.7	2,065.1
C02	Textile & Leather Products	4,997.8	4,964.9	5,350.2	4,960.9	4,608.8	4,146.2	4,368.2	4,233.5	4,524.8
C03	Wood, Paper & Printing	1,092.5	2,334.5	2,420.3	2,455.8	2,560.9	2,523.4	2,504.5	2,812.9	2,843.8
C04	Coal & Petroleum Products	35,181.4	35,650.3	33,332	34,774.4	36,382	33,423.9	33,953.5	34,296.9	36,281.6
C05	Chemical Products	62,139.9	68,515.1	73,030.3	77,404.7	76,680	78,677.3	87,060.5	84,691.7	84,687.5
C06	Non-metallic Mineral Products	42,987.5	43,823.5	42,638.7	42,972.8	40,522.2	37,377.5	39,163.2	37,524.1	35,157.3
C07	Primary Metal Products	113,228	106,984.1	111,438.4	113,154.4	111,377.7	106,174.3	109,192.3	100,331.6	102,537.7
C08	Fabricated Metal Products	1,237.4	1,655.8	2,016.5	843.7	938.1	418	415.5	363	368.3
C09	Computer, Electronic & Optical Equipment	8,962.5	6,561.8	8,382.6	9,118.4	8,314.3	8,837.7	9,055.5	7,078.9	3,829.2
C10	Electrical Equipment	631.4	886	1,132.7	392.2	432.6	130.9	144.1	126.7	123.9
C11	Machinery & Equipment	3,034.7	3,654.6	4,814.1	4,594.7	5,319.1	5,465.4	5,802.1	6,083.1	6,927
C12	Transport Equipment	2,330.5	3,674.2	3,468.4	2,446.9	2,215.4	2,190.3	2,254.9	2,240.4	2,362.2
C13	Other Manufacturing Products	2,991.2	3,658.9	3,437.7	2,936.9	4,496	3,710.4	4,231.6	3,305.3	3,595.1
C14	Manufacturing Processing & Industrial Equipment Repair	1,083.3	1,127.5	1,155.5	950	837.1	1,694.9	1,648.2	1,646.1	1,648.6
D	Electricity, Gas & Steam	248,052	248,366.4	258,935.4	279,923.9	260,702.2	229,935.1	242,672.3	240,926.7	229,332
E	Water, Waste Management & Recycling Services	19,752.3	19,628.8	20,151	19,336.6	18,747.3	18,461.3	18,121.8	18,121.8	18,172
F	Construction	2,611	2,686.2	2,261.2	2,218.1	2,183.4	2,139	2,034.9	1,966.8	1,813
G	Wholesale, Retail & Commission Trade Services	2,075.6	2,087.5	2,148.1	1,792	1,498	1,756.9	1,676.6	1,519.2	1,433.1
H	Transportation Services	93,028.6	101,457	102,846.4	101,215.9	98,036.7	91,742.5	95,090.8	97,490.4	100,529.1
I	Food & Accommodation Services	2,389.9	3,015	3,102.6	3,067	2,909.5	2,231.4	2,184.9	2,266.4	2,253.1
J	Information, Communication & Broadcasting Services	210.7	212.7	218.9	188	155.2	96.1	96.1	105.4	101.4
K	Financial & Insurance Services	215	206.8	212.8	196.9	175.6	123.5	127.7	133	132
L	Real Estate Services	429.8	516.7	531.7	472.4	422	246.6	251.8	249.6	236.2
M	Professional, Scientific & Technical Services	2,178.1	2,309.2	2,376.2	2,289	1,982	1,667.8	1,663.2	1,779.6	1,892.7
N	Business Support Services	293.7	306.9	315.8	284.5	241	141.4	135.1	151.6	157.6
O	Public Administration, Defense & Social Security	6,946.6	6,889.1	7,104.5	6,942	6,317.4	5,829.2	5,866.9	6,004.8	6,015.9
P	Education Services	2,593.8	2,266.6	2,332.4	2,061.7	1,912.2	2,108.8	2,123.9	2,197.9	2,189.5
Q	Health & Social Welfare Services	2,334.4	2,568.5	2,643.1	2,700.3	2,606.1	1,698.1	1,710.2	1,772.3	1,803.2
R	Arts, Sports & Leisure-related Services	529.3	571.7	588.3	584.2	495.2	344.2	343.5	369.5	373.4
S	Other Services	788.1	687.6	707.5	670.1	562.9	426.5	413.4	442.8	422.1
T	Other	0	0	0	0	0	0	0	0	0
9090	Intermediate Demand Total	695,797.6	708,333.5	730,078.1	751,669.5	723,871.8	674,365.8	704,550.5	690,151.1	682,059.5
9111	Private Consumption Expenditure	70,852.6	73,946.3	73,831	75,212.8	75,931.2	77,381.3	76,878.6	78,768.4	75,637.9
	Grand Total	766,650.2	782,279.8	803,909	826,882.4	799,803	751,747.2	781,429	768,919.5	757,697.5

1.3 Constant-Price Industry-Linkage Table

- Time-series analysis requires a constant-price industry-linkage table; in particular, constant-price output and the constant-price domestic transaction table are used to calculate various coefficients such as the GHG emission coefficient



2. Carbon Inducement Coefficient Calculation Pipeline

Core logic for carbon inducement coefficient calculation — Leontief model based on the domestic transaction table

1

Input Coefficient Matrix A

Divide the domestic transaction table Z by the total input vector X to quantify the domestic input structure between industries

$$A = Z / X$$

2

Leontief Inverse Matrix L

Derive the inverse matrix that captures all direct and indirect supply-chain ripple effects between industries

$$L = (I - A)^{-1}$$

3

Direct Emission Coefficient B

Unit emission coefficient obtained by dividing GHG emissions by industry (E, kt→t) by total input (X)

$$B = E / X$$

4

Supply-Chain Inducement Coefficient B×L

Multiply the direct emission coefficient by the Leontief inverse matrix to derive the total supply-chain inducement effect

$$BL = B \cdot L, \quad M = \text{diag}(B) \cdot L$$

* Before calculation, verify that the industry code order matches between the total and domestic transaction tables, and confirm the presence of the D(electricity) code

2.1 Scope 1·2·3 Decomposition Logic

Corrects for double-counting in the electricity (D) sector by splitting emissions into three scopes

Code logic (eeio_core_final.py)

```
scope_1 = B.copy()

scope_2_raw = M[D_row, :].copy()
scope_2_raw[D_row] -= scope_1[D_row]
scope_2 = max(scope_2_raw, 0)

scope_3_raw = BL - scope_1 - scope_2
scope_3 = max(scope_3_raw, 0)
```

Scope 1 Direct Emissions

= B (direct process/combustion emissions of the industry itself)

Scope 2 Indirect Energy Emissions

= electricity (D) row value of the M matrix – electricity's own direct emissions (extracts only the indirect portion induced through electricity consumption)

Scope 3 Indirect Supply-Chain Emissions

= $B \times L$ (supply-chain inducement coefficient) – Scope1 – Scope2 (remaining supply-chain induced portion from raw material/intermediate goods procurement, etc.)

* D_rel(electricity sector position): why its own direct emissions (B) are subtracted from the M-matrix row value — M[D,:] already includes electricity's own direct emissions, so they must be excluded to leave only the "indirect emissions induced by other industries consuming electricity."

3. Analysis Results and Model hub Upload

Scope Summary Dashboard

Korea Industry-Level Scope 1, 2, 3 Carbon Emissions Dashboard (2023, current prices)

All 33 industries | sorted by total emissions descending | Upstream/Downstream based on the M matrix

Korea Industry-wide Scope 1, 2, 3 Carbon Emissions Comprehensive Dashboard (2023, Current Prices)

All 32 industries | Sorted by total emissions in descending order | Upstream/Downstream based on carbon-induced (M matrix)

Comprehensive Insight – Carbon Emissions Structure Analysis Across All Industries

- The combined carbon footprint of all 32 industries is 1,612,503,745.6925 t CO₂eq, of which Scope 1 (Direct) accounts for 42.3%, Scope 2 (Energy) 24.5%, and Scope 3 (Supply chain) 33.2%.
- The top 3 industries by total emissions (Electricity, gas and steam; Primary metal products; Chemical products) account for 38.8% of the entire supply chain carbon footprint, indicating that emissions are concentrated in a small number of key industries.
- The industry with the highest Scope 1 share is 'Electricity, gas and steam' (81%), where managing direct emissions from processes and combustion is urgent, while the industry with the highest Scope 3 share is 'Construction' (76%), where supply chain management is key to carbon reduction.

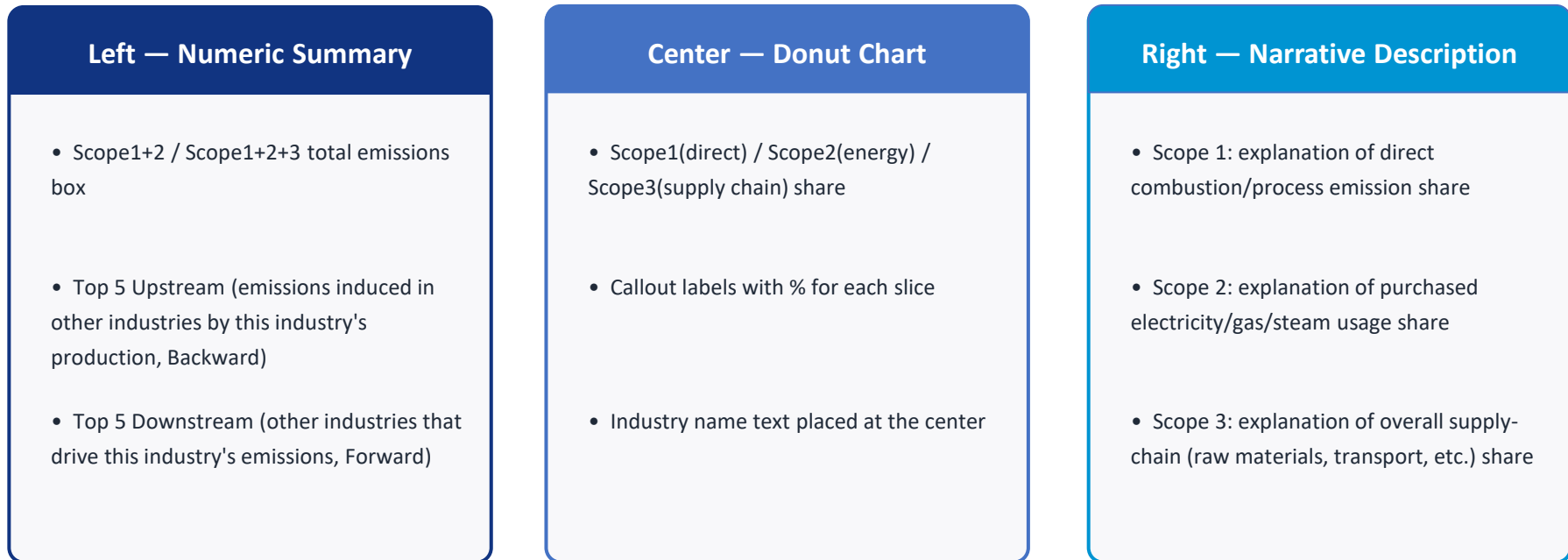
- Aggregate carbon footprint and Scope 1/2/3 share across all industries · Concentration (share %) of the top 3 emitting industries
- Auto-highlight of the industry with the highest Scope 1/3 share · Narrative on the need for supply-chain decarbonization for industries with a high Scope 3 share

3. Analysis Results and Model hub Upload

Scope Dashboard — Industry Detail Cards

One card per industry, 3-column layout

Left — numeric summary | Center — donut chart | Right — narrative description



** Upstream/Downstream values are the top 5 extracted based on the column/row vector for the industry from the M matrix ($\text{diag}(B) \cdot L$, $\times 1000$ scale)*

3. Analysis Results and Model hub Upload

Scope Dashboard — Industry Detail Cards

#1 Electricity, Gas and Steam

Korea EEIO Analysis Results | 2023 Current Prices

Scope 1+2	Scope 1+2+3 (Total Emissions)
278,608,881.2666	283,078,134.7832

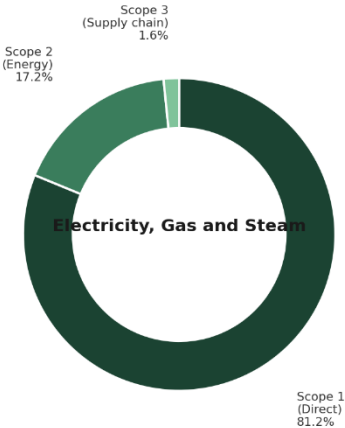
Unit: t CO₂eq / KRW million (current prices)

Top 5 Upstream (Actual emissions induced in other industries by this industry's production, t CO₂eq)

1. Chemical products	1,112,241.8048
2. Transportation services	1,009,692.4871
3. Primary metal products	796,758.9557
4. Coal and petroleum products	788,998.3113
5. Water supply, waste treatment and recycling services	285,531.1484

Top 5 Downstream (Actual emissions induced in other industries' production by this industry's emissions, t CO₂eq)

1. Chemical products	38,263,186.8123
2. Primary metal products	33,549,952.4902
3. Computer, electronic and optical equipment	28,398,737.1048
4. Transportation equipment	27,534,888.0051



Scope 1, 2, 3 Emissions Structure

Electricity, Gas and Steam

Scope 1 (81%)

Direct combustion and process emissions account for 81% of this industry's emissions.

Scope 2 (17%)

Indirect emissions from purchased electricity, gas, and steam use account for 17%.

Scope 3 (2%)

The remaining 2% is induced across the entire supply chain (raw materials, transportation, etc.).

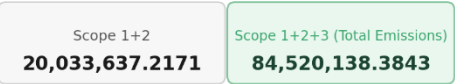
* Approximate figures based on aggregated industry data, current prices

3. Analysis Results and Model hub Upload

Scope Dashboard — Industry Detail Cards

#5 Construction

Korea EEIO Analysis Results | 2023 Current Prices



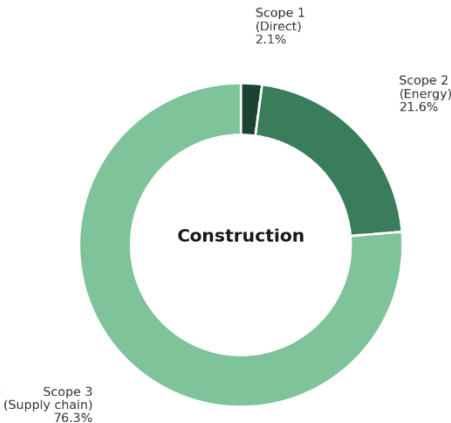
Unit: t CO₂eq / KRW million (current prices)

Top 5 Upstream (Actual emissions induced in other industries by this industry's production, t CO₂eq)

1. Non-metallic mineral products	22,680,938.3685
2. Primary metal products	21,254,008.5085
3. Electricity, gas and steam	18,220,617.2171
4. Transportation services	7,473,435.8532
5. Chemical products	5,997,150.8339

Top 5 Downstream (Actual emissions induced in other industries' production by this industry's emissions, t CO₂eq)

1. Real estate services	104,150.2289
2. Wholesale, retail and commodity brokerage services	12,993.4767
3. Professional, scientific and technical services	11,324.5978
4. Restaurant and accommodation services	11,096.5886



Scope 1, 2, 3 Emissions Structure

Construction

Scope 1 (2%)

Direct combustion and process emissions account for 2% of this industry's emissions.

Scope 2 (22%)

Indirect emissions from purchased electricity, gas, and steam use account for 22%.

Scope 3 (76%)

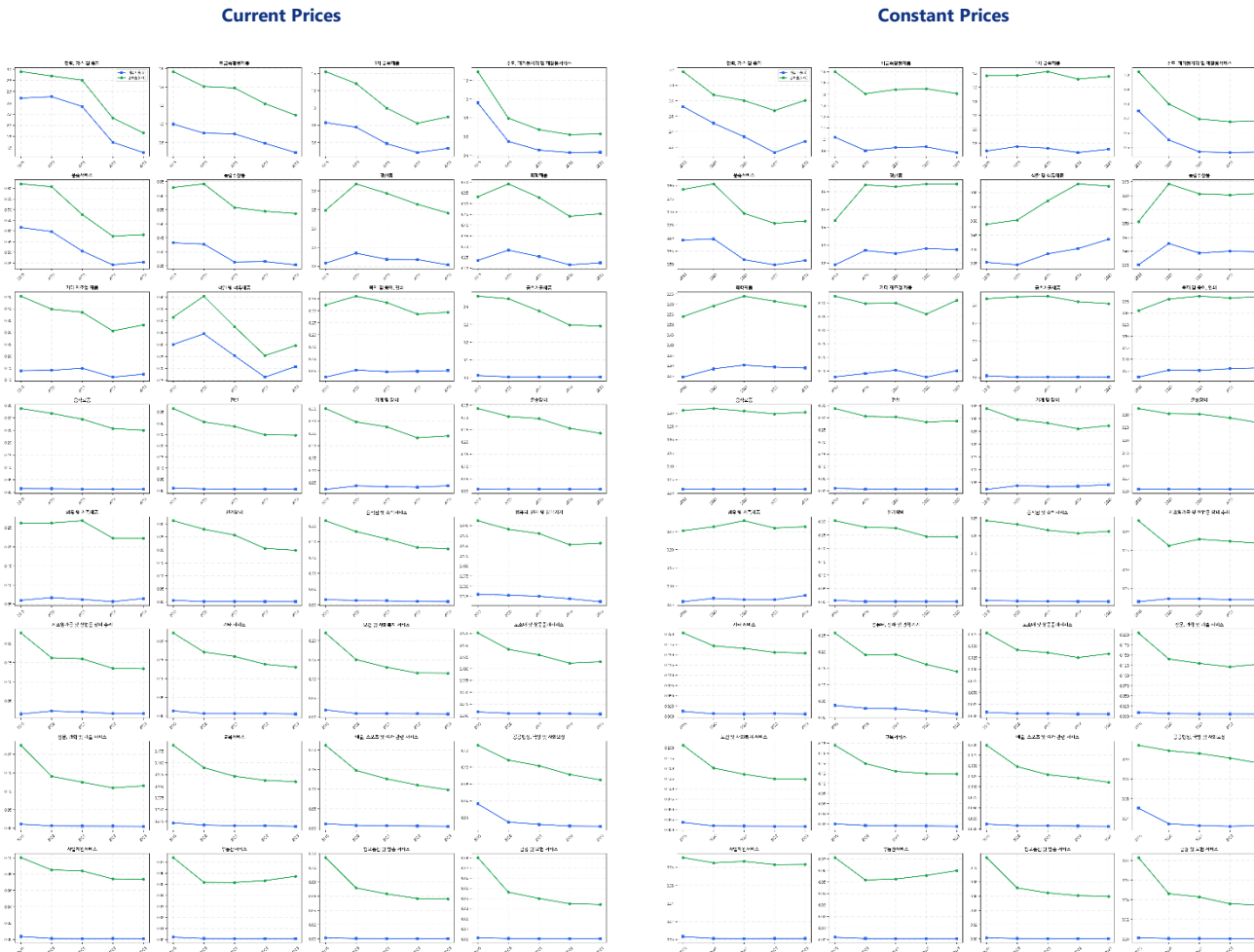
The remaining 76% is induced across the entire supply chain (raw materials, transportation, etc.).

* Approximate figures based on aggregated industry data, current prices

3. Analysis Results and Model hub Upload

Direct Emissions(B) vs Total Emissions(B×L) Time Series

All industries (33) arranged in a 4-column grid, with both coefficients overlaid by year in each cell



How to Read the Chart

■ Blue line — direct emission coefficient (B)

■ Green line — supply-chain inducement coefficient (B×L)

Gap between the two lines = indirect induction via the supply chain (B×L – B)

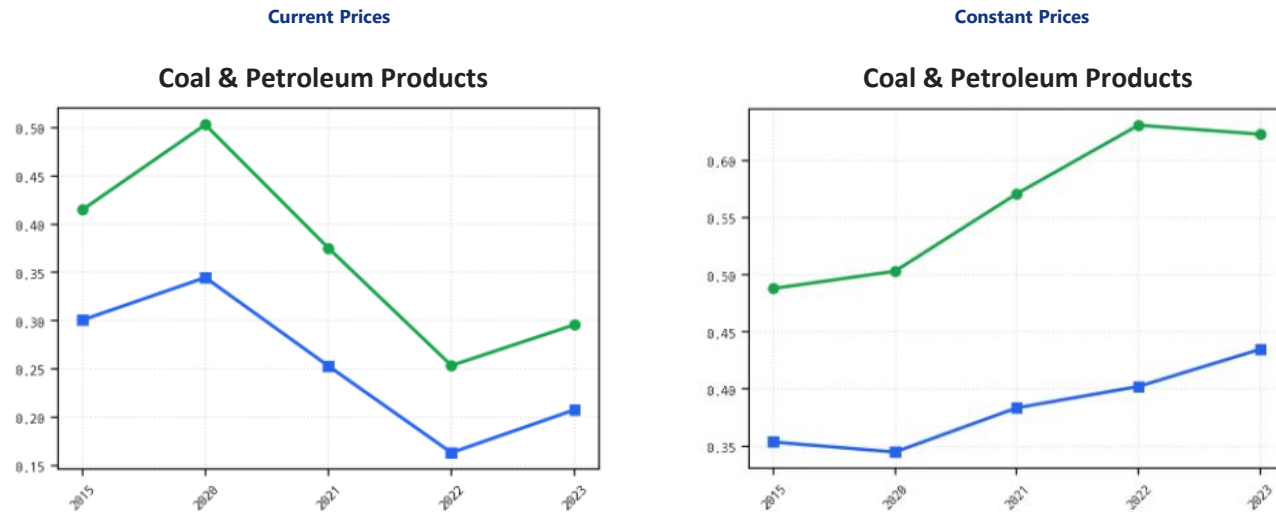
A larger gap indicates an industry with higher supply-chain-driven emissions

Year axis: 2015, 2020~2023

3. Analysis Results and Model hub Upload

Direct Emissions(B) vs Total Emissions($B \times L$) Time Series

All industries (33) arranged in a 4-column grid, with both coefficients overlaid by year in each cell



Coal and Petroleum Products

- Current-price and constant-price patterns can be seen to differ
- After removing the price effect, the direct emission coefficient and the supply-chain inducement coefficient turn to an increasing trend
- In constant prices, the gap in indirect induction ($B \times L - B$) widens, showing markedly higher supply-chain-driven emissions

4. GHG-Simulator

- Enables SMEs and others to estimate total emissions and Scope 1·2·3 using KR-EEIO

Add company information to the already-compiled KR-EEIO(33x33)
(matching industry classification, 34x34)

- **Calculation pipeline**
 - $A \rightarrow L \rightarrow B \rightarrow B \times L$ calculation
 - Scope 1·2·3 decomposition logic (electricity double-counting correction)
 - Calculation logic and outputs of the corporate GHG-Simulator

4. GHG-Simulator

Enter a company's sales and cost structure to automatically estimate Scope 1-2-3 based on 2023 KR-EEIO (current price) data

STEP 1

Company Name

COMPANY_NAME

Used in the result file name

STEP 2

Industry Code

COMPANY_CODE

Select from the 33-industry code table
(e.g., C05 Chemical products)

STEP 3

Sales

SALES

Unit: KRW million

STEP 4

Cost / Value-Added Share

COST_RATIOS / VA_RATIOS

Unknown items = 0 → auto-estimated from industry ratios

3 calculation modes depending on input completeness

Case 1

Only sales known

Leaving all cost items at 0 applies the national-average industry structure for estimation

Case 2

Some cost items known

Enter known items only; the rest is auto-filled with national averages (example run)

Case 3

All costs known

Enter directly so that COST_RATIOS + VA_RATIOS sum to 100%

Calculation method is identical to the existing EEIO pipeline — Scope1 from sales × direct emission coefficient (B), Scope2 from the electricity-row correction of the M matrix ($\text{diag}(B) \cdot L$), and the remaining supply-chain induction as Scope3

4. GHG-Simulator

- SMEs and others, using KR-EEIO, estimate total emissions and Scope1, 2, 3

Chemical products(C05) For a company in the industry with sales of **1000**, and cost shares of Wood, paper and printing (C03) **5%**, Coal and petroleum products(C04) **10%** entered

< Total Transaction Table >

Product		A	B	C01	C02	C03	C04	C05	C06	C07	C08	
		A Agriculture, B Mining Prod C01 Food & Be C02 Textile & L C03 Wood, Pa C04 Coal & Pe C05 Chemical							Analyzed Firms C06 Non-met	C07 Primary M	C08 Fabricated Met	
A	Agriculture, For	4256311	3473	44787107	264444	661849	4203	1326554.295	4.704636154	3614	6461	7383
B	Mining Product	2055	2999	48702	3592	35023	108067377	8679176.219	30.78077269	5208178	33232955	40482
C01	Food & Beverage	13490896	16188	37267582	1267831	87762	29747	1948947.088	6.911957516	32354	78230	106035
C02	Textile & Leather	690669	18761	152583	21941986	610886	100196	1674114.063	5.937259841	167311	181532	388657
C03	Wood, Paper & Printing	866017	11121	3715108	415547	17244503	134210	1997325	50	426375	243459	1061021
C04	Coal & Petroleum	1553720	127562	368746	182073	306192	8485744	43895147	100	2409467	721361	527953
C05	Chemical Products	5335744.955	219297.5461	4720436.229	6819714.883	3803162.128	3537787.677	129362110.4	458.7847807	2543479.735	3102888.577	9009071.352
	Analyzed Firms	11.04475912	0.453936347	9.771096918	14.11651209	7.872379573	7.323066045	267.7748708	0	5.264892013	6.422843895	18.6483844
C06	Non-metallic Mineral	36617	8918	905256	41584	59557	153911	1621299.25	5.749951656	7622593	3265699	612991
C07	Primary Metal	91111	8196	26239	2697	45720	562107	3388699.982	12.01805347	903550	62564642	26175296
C08	Fabricated Metal	77940	117861	2371080	248175	609407	2900135	2673611.518	9.481986114	667203	1573739	20929812

< Domestic Transaction Table >

Product		A	B	C01	C02	C03	C04	C05	C06	C07	C08	
		Agriculture, For Mining Product Food & Beverage Textile & Leather Wood, Paper & Printing Coal & Petroleum Chemical Products Analyzed Firms Non-metallic Mineral Primary Metal Fabricated Metal										
A	Agriculture, Forestry & Fisheries Products	3770505	3469	34517696	38677	277682	4203	555661.0293	1.970656593	3614	6461	7383
B	Mining Products	2055	2999	41769	2547	11840	112216	373201.6764	1.323562937	3949468	229276	35343
C01	Food & Beverage Products	13273857	16188	23135036	86468	67764	-9926	1348686.217	4.783127202	32354	78230	106034
C02	Textile & Leather Products	516322	18761	131097	9447326	469462	99962	1070188.205	3.79543162	150174	178752	379901
C03	Wood, Paper & Printing	858139	11121	3523836	218583	12285532	133424	1854725.57	46.43023969	387612	234487	982943
C04	Coal & Petroleum Products	1515963	125960	326179	113399	238255	4806990	18548271.74	42.25585973	2257768	507483	449612
C05	Chemical Products	4512613.955	217551.5461	4252088.229	5435480.883	3512885.128	2835504.677	75442644.67	267.5589342	1751714.735	1645878.577	7862773.352
	Analyzed Firms	11.04475912	0.453936347	9.771096918	14.11651209	7.872379573	7.323066045	267.7748708	0	5.264892013	6.422843895	18.6483844
C06	Non-metallic Mineral Products	36349	8909	866923	29958	57935	128152	1034796.33	3.669914034	6026519	2828197	509833
C07	Primary Metal Products	90703	8189	26239	-2016	45297	548043	1432931.918	5.081905302	880571	33775453	21994525
C08	Fabricated Metal Products	76325	112396	2217143	116512	594802	2872386	2326397.749	8.250589514	586542	1481015	18262217

< Calculation Results >

Analyzed firm	BOK firm
Industry	C05
Sales(1MW)	1,000
Calculation mode	Case2
Scope1 (tCO2eq.)	224.9
Scope2 (tCO2eq.)	139.3
Scope3 (tCO2eq.)	170.0
Total CO2 emission (tCO2eq.)	534.2

4. GHG-Simulator

- Example Run Results

Company Information

Company Name	BOK Company
Industry	C05 Chemical products
Sales (KRW million)	1,000
Calculation Mode	Case2 (some costs known)

Carbon Emission Results — Scope 1 · 2 · 3

Scope1 (Direct emissions)	224.9 tCO2eq	42.1%
Scope2 (Electricity, etc. indirect emissions)	139.3 tCO2eq	26.1%
Scope3 (Other indirect emissions)	170.0 tCO2eq	31.8%
Total (Scope 1+2+3)	534.2 tCO2eq	100.0%

* Example input: COMPANY_CODE=C05, SALES=1,000 KRW million, only wood/petroleum-product costs partially entered (Case2)

Emission Contribution by Industry (Top 8)

1	Company analyzed	224.9	42.1%
2	D · Electricity, gas and steam	139.3	26.1%
3	C05 · Chemical products	81.3	15.2%
4	H · Transport services	36.1	6.8%
5	C04 · Coal and petroleum products	15.5	2.9%
6	E · Water · waste treatment · recycling	11.7	2.2%
7	C07 · Primary metal products	9.2	1.7%
8	C06 · Non-metallic mineral products	4.8	0.9%

5. Summary and Future Tasks

Summary

Methodology Summary

Calculated in four steps — domestic input coefficient(A_d) → Leontief inverse matrix(L_f) → emission coefficient(B) → supply-chain inducement coefficient(M) in 4 steps, with the electricity(D) sector separately corrected to split Scope 1/2/3.

Visualization and Output Summary

① Scope dashboard, ② B vs $B \times L$ grid time series, ③ GHG-Simulator result tables can all be viewed directly in the Model hub notebooks.

Notes

Price-Basis Dependency

Current prices are for practical calculations, while constant prices (2020=100) serve a different purpose — pure volume comparison across years.

Simulator Estimate Limitations

Unentered cost items are replaced with industry averages, so error can grow larger if the actual cost structure differs.

Model hub Contents

Source Code

- eeio_core_final.py
- eeio_simulator.py

Jupyter Notebooks

- KR-EEIO_result.ipynb
- KR-EEIO_Simulator.ipynb

5. Summary and Future Tasks

Future Tasks

Acquiring official data from Input-Output Team

Replacing current emission estimates with emission data to be officially estimated by the Input-Output Team

Incorporate Import Analysis

Add analysis of imported goods in addition to domestically produced goods (GHG emissions analysis by importing country using international industry-linkage tables, etc.)

Compile Based on Basic Prices

The current KR-EEIO, based on an industry-linkage table published in producer prices, KR-EEIO includes net taxes on products, so it should be compiled based on basic prices with each industry's tax effect removed — KR-EEIO compilation

Expand Sector Count & Environmental Statistics

-Expanded to the basic-sector level (380 industries) based on GHG emissions by energy source and other data, increasing KR-EEIO's utility

- 7 individual GHGs' emissions, energy, waste, and wastewater/water — by incorporating these and other environmental statistics, KR-EEIO's coverage will be progressively expanded

Thank You